

Metrics and Ratios Reference

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Personal finance ratios

Net worth

- Used for: measuring financial position at one date
- Formula: Total assets – Total liabilities

Net cash flow

- Used for: seeing whether income covers spending over a period
- Formula: Income – Expenses

Savings ratio

- Used for: checking how much of gross income is saved (benchmark 10% in Lecture 2S; 20% or more in the videos and Lectures 3 and 7)
- Formula: Total savings ÷ Gross income

Debt servicing ratio

- Used for: checking whether debt payments are manageable (benchmark under 36%)
- Formula: Total debt payments ÷ Gross income

Debt ratio

- Used for: checking how leveraged the household is (benchmark under 40%)
- Formula: Total liabilities ÷ Total assets

Current ratio

- Used for: checking short-term liquidity
- Formula: Cash ÷ Short-term liabilities

Housing ratio

- Used for: checking whether housing costs are affordable (benchmark 28% or less)
- Formula: Housing costs ÷ Gross pay

Emergency fund

- Used for: sizing the cash cushion for job loss or surprise costs

- Formula: $3 \text{ to } 6 \times \text{Monthly necessary expenses}$

50-30-20 budget

- Used for: splitting after-tax income into needs, wants and savings or debt repayment
- Formula: Needs 50% + Wants 30% + Savings and debt repayment 20% of monthly after-tax income

Down payment and mortgage

- Used for: sizing a home purchase with a 20% down payment
- Formula: Down payment = Price $\times$ 20%; Mortgage = Price – Down payment

Annual interest on savings

- Used for: estimating yearly interest, and the gain from moving to a better account
- Formula: Balance $\times$ Rate; Gain from switching = Balance $\times$ (New rate – Old rate)

Tax measures

Gross income

- Used for: the starting point of the tax return
- Formula: Income – Exclusions

Adjusted gross income (AGI)

- Used for: the base for many limits and phase-outs
- Formula: Gross income – Deductions for AGI (adjustments)

Taxable income

- Used for: the amount the tax brackets apply to
- Formula: AGI – max(Standard deduction, Itemized deductions) – QBI deduction

Tax due or refund

- Used for: the final amount owed or returned
- Formula: Tax on taxable income – Credits; Refund = Federal tax withheld – (Income tax – Credits)

Effective tax rate

- Used for: the average rate actually paid
- Formula: Total income tax $\div$ Taxable income

Marginal tax rate

- Used for: the rate on the next dollar earned or deducted
- Formula: The bracket rate that applies to the last dollar of taxable income

Deduction savings and credit savings

- Used for: comparing a deduction (worth less than face value) to a credit (dollar for dollar)
- Formula: Deduction savings = Deduction $\times$ Marginal rate; Credit savings = Credit amount

Taxes on earnings

- Used for: combined state and federal tax on pretax earnings
- Formula: Pretax earnings $\times$ (State rate + Federal rate)

Capital gain or loss

- Used for: the taxable result of selling an investment (long term if held more than one year)
- Formula: Proceeds – Cost basis

Total tax rate

- Used for: the combined rate applied to investment income
- Formula: Federal marginal rate + State rate

After-tax yield

- Used for: comparing taxable and tax-free bonds on equal footing
- Formula: Pre-tax yield $\times$ (1 – Total tax rate)

Tax-equivalent yield

- Used for: expressing a tax-free muni yield as a taxable yield
- Formula: Tax-free yield $\div$ (1 – Tax rate)

Retirement and growth

4% rule portfolio

- Used for: sizing the portfolio needed to fund retirement spending
- Formula: Annual retirement spending $\div$ 4%

Inflated spending

- Used for: converting today's spending into dollars at retirement
- Formula: $\text{Spending} \times (1 + \text{inflation})^n$

Portfolio needed at retirement

- Used for: applying the 4% rule to inflated spending
- Formula: $\text{Spending} \times (1 + \text{inflation})^n \div 4\%$

Future value

- Used for: growing a lump sum at a fixed return
- Formula: $PV \times (1 + r)^n$

Required return

- Used for: finding the annual return needed to reach a goal
- Formula: $(FV \div PV)^{(1/n)} - 1$

Return measures**Arithmetic average return**

- Used for: a simple average of periodic returns; ignores compounding
- Formula: $\sum r_t \div n$

Geometric average return

- Used for: the compound rate actually earned (for example +25% then -20% is 0%, not 2.5%)
- Formula: $[\prod (1 + r_t)]^{(1/n)} - 1$

Real return

- Used for: return after removing inflation
- Formula: $(1 + \text{Nominal return}) \div (1 + \text{Inflation}) - 1$

Portfolio expected return

- Used for: the weighted average return of the holdings
- Formula: $E(R_p) = \sum w_i \times R_i$

Realized stock return

- Used for: explaining where a stock's return came from

- Formula: Dividend yield + Dividend growth + Change in the multiple

Risk measures

Standard deviation

- Used for: total volatility of returns
- Formula: $\sigma = \sqrt{\sum (x_i - \bar{x})^2 \div (n - 1)}$

Coefficient of variation

- Used for: risk per unit of return, to compare investments with different averages
- Formula: $CV = \sigma \div \text{Mean}$

Correlation

- Used for: how two returns move together, from -1 to +1; drives diversification
- Formula: $\rho = \text{Cov}(x, y) \div (\sigma_x \times \sigma_y)$

Beta

- Used for: sensitivity to the market (1.5 moves 1.5 times the market)
- Formula: $\beta = \text{Cov}(R_p, R_m) \div \sigma_m^2$

Portfolio beta

- Used for: market sensitivity of a whole portfolio
- Formula: $\beta_p = \sum w_i \times \beta_i$

Tracking error

- Used for: how far a portfolio's returns stray from its benchmark
- Formula: $\sigma(R_p - R_b)$

Risk-adjusted performance

CAPM expected return

- Used for: the return required for the market risk taken
- Formula: $E(R_i) = R_f + \beta_i \times (R_m - R_f)$

Alpha

- Used for: return beyond what beta predicts (positive means outperformance)

- Formula: $\alpha = R_{\text{actual}} - E(R) = R_p - [R_f + \beta_p \times (R_m - R_f)]$

Sharpe ratio

- Used for: excess return per unit of total risk (below 1 weak, above 1 good, above 2 very good, 3 or more excellent)
- Formula: $(R_p - R_f) \div \sigma_p$

Treynor ratio

- Used for: excess return per unit of market risk
- Formula: $(R_p - R_f) \div \beta_p$

Information ratio

- Used for: active return earned per unit of tracking error
- Formula: $(R_p - R_b) \div \text{Tracking error}$

Allocation effect

- Used for: attribution of return to over- or underweighting sectors
- Formula: $(w_p - w_b) \times (R_{b,\text{sector}} - R_{b,\text{total}})$

Selection effect

- Used for: attribution of return to picking better securities within a sector
- Formula: $w_b \times (R_p - R_b)$

Interaction effect

- Used for: attribution of return to the combination of allocation and selection
- Formula: $(w_p - w_b) \times (R_p - R_b)$

Valuation

Price-to-earnings (P/E)

- Used for: the price paid per dollar of earnings (trailing or forward)
- Formula: $\text{Price} \div \text{Earnings per share}$

CAPE (cyclically adjusted P/E)

- Used for: valuing the market against 10 years of inflation-adjusted earnings, smoothing the business cycle
- Formula: $\text{Price} \div \text{Average inflation-adjusted earnings over the past 10 years}$

Dividend yield

- Used for: the cash return from dividends
- Formula: Next-12-month dividend $\div$ Price

Earnings yield

- Used for: the inverse of P/E, for comparing stocks with bond yields
- Formula: $\text{EPS} \div \text{Price} = 1 \div (\text{P/E})$

Earnings yield spread

- Used for: stocks versus corporate bonds (negative suggests expensive equities)
- Formula: Earnings yield – Baa yield

Gordon equation

- Used for: a rough expected stock return from dividend yield plus growth
- Formula: $E(R) \approx D \div P + g$